

Laplace Transforms

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1 Definition

If a function is just a set of instructions that take in an input number and spits out an output number, Laplace transforms take in a function and spits out another function.

The Laplace transform that transforms a function f in t to another function in s is denoted by $\mathcal{L}\{f(t)\}(s) = F(s)$. Mathematically,

$$\mathcal{L}\{f(t)\}(s) = \int_0^{\infty} f(t)e^{-st} dt$$

2 Important Properties

2.1 Laplace Transforms of simple functions:

$$\mathcal{L}\{c\}(s) = \frac{c}{s} \quad (\text{where } c \text{ is a constant})$$

$$\mathcal{L}\{t\}(s) = \frac{1}{s^2}$$

$$\mathcal{L}\{t^2\}(s) = \frac{2}{s^3}$$

$$\mathcal{L}\{t^n\}(s) = \frac{n!}{s^{n+1}}$$

$$\mathcal{L}\{\sin(at)\}(s) = \frac{a}{s^2 + a^2}$$

$$\mathcal{L}\{\cos(at)\}(s) = \frac{s}{s^2 + a^2}$$

$$\mathcal{L}\{e^{-at}\}(s) = \frac{1}{s + a}$$

Algebraic Properties

$$\mathcal{L}\{a \cdot f + b \cdot g\} = a \mathcal{L}\{f\} + b \mathcal{L}\{g\}$$

Scaling Property : if $\mathcal{L}\{f(t)\}(s) = F(s)$, then $\mathcal{L}\{f(ct)\}(s) = \frac{1}{c}F(\frac{s}{c})$

2.2 Convolution of two functions

Convolution is just another operation that acts between two functions just like addition, subtraction, multiplication of two functions. It is special because it preserves the property of the individual functions and while taking the Laplace transform of a convolved function, it becomes product of the transformed functions

$$\mathcal{L}\{f * g\} = \mathcal{L}\{f\} \cdot \mathcal{L}\{g\}$$

This implies, convolution is commutative $\implies f * g = g * f$. Convolution of two functions $f(t)$ and $g(t)$ is defined as

$$(f * g)(t) = \int_0^t f(t - \tau) g(\tau) d\tau = \int_0^t f(\tau) g(t - \tau) d\tau$$

2.3 Shifting properties

The First shifting property (s-domain shifting)

if $\mathcal{L}\{f(t)\}(s) = F(s)$,

$$\boxed{\mathcal{L}\{e^{at}f(t)\}(s) = F(s - a)}$$

It means that, by multiplying the function f by e^{at} , I am essentially shifting the Laplace transformed function a units to the right. That is, $F(s)$ becomes $F(s - a)$

The Second shifting property (t-domain shifting)

if $\mathcal{L}\{f(t)\}(s) = F(s)$,

$$\boxed{\mathcal{L}\{u_a f(t - a)\}(s) = e^{-as} F(s)}$$

It means that, by multiplying the function $F(s)$ by e^{-as} , I am essentially shifting the inverse Laplace transformed function a units to the right. That is, $f(t)$ becomes $f(t - a)$.

$$\begin{aligned} \mathcal{L}\{u_a(t) \cdot f(t - a)\}(s) &= \int_0^\infty u_a(t) \cdot f(t - a) e^{-st} dt \\ &= \int_0^a u_a(t) \cdot f(t - a) e^{-st} dt + \int_a^\infty u_a(t) \cdot f(t - a) e^{-st} dt \\ &= \int_0^a 0 \cdot f(t - a) e^{-st} dt + \int_a^\infty 1 \cdot f(t - a) e^{-st} dt \\ &= 0 + \int_a^\infty f(t - a) e^{-st} dt \end{aligned}$$

This is why the Heaviside(at a) function was important here. Without it, the $\int_0^a$ part won't be equal to zero. As we shall see soon, this part being zero is necessary for us to get the final simplified integral as $\int_0^\infty$ so that we can write it in Laplace format. For all this to happen, we must first substitute $t - a = \tau \implies dt = d\tau$

$$\begin{aligned}
&= \int_0^\infty f(\tau) e^{-s(\tau+a)} d\tau \\
&= e^{-as} \int_0^\infty f(\tau) e^{-s\tau} d\tau \\
&= e^{-as} \int_0^\infty f(t) e^{-st} dt \\
&= e^{-as} \mathcal{L}\{f(t)\}(s)
\end{aligned}$$

Laplace transform assumes that nothing happened before $t = 0$, The Heaviside function ensures that the function takes the value 0 for all $t < a$.

2.4 Laplace of an Integral

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{1}{s} \mathcal{L}\{f\}(s)$$

proof :

$$\text{let } g(t) = 1 \implies g(t - \tau) = 1$$

$$\begin{aligned}
\mathcal{L}\left\{\int_0^t 1 \cdot f(\tau) d\tau\right\} &= \mathcal{L}\left\{\int_0^t g(t - \tau)f(\tau) d\tau\right\} \\
&= \mathcal{L}\{g * f\} \\
&= \mathcal{L}\{g\} \mathcal{L}\{f\} \\
&= \mathcal{L}\{1\} \mathcal{L}\{f\} \\
&= \frac{1}{s} \mathcal{L}\{f\}(s)
\end{aligned}$$

2.5 Laplace of a Derivative

$$\mathcal{L}\{f'(t)\}(s) = s \cdot \mathcal{L}\{f(t)\}(s) - f(0)$$

proof :

According to the fundamental theorem of calculus,

$$\begin{aligned} \int_a^b f'(t) d\tau &= f(b) - f(a) \\ \implies \int_0^t f'(\tau) d\tau &= f(t) - f(0) \end{aligned}$$

Taking Laplace on both sides,

$$\begin{aligned} \mathcal{L} \left\{ \int_0^t f'(\tau) d\tau \right\} &= \mathcal{L} \{f(t) - f(0)\} \\ \implies \frac{1}{s} \mathcal{L} \{f'(t)\} (s) &= \mathcal{L} \{f(t)\} - \mathcal{L} \{f(0)\} \\ \implies \frac{1}{s} \mathcal{L} \{f'(t)\} (s) &= \mathcal{L} \{f(t)\} - \frac{f(0)}{s} \\ \mathcal{L} \{f'(t)\} (s) &= s \cdot \mathcal{L} \{f(t)\} (s) - f(0) \end{aligned}$$

2.6 Laplace of a Double Derivative

$$\boxed{\mathcal{L}\{f''(t)\}(s) = s^2 \cdot \mathcal{L}\{f(t)\}(s) - s \cdot f(0) - f'(0)}$$

proof :

According to the fundamental theorem of calculus,

$$\begin{aligned} \int_a^b f''(t) d\tau &= f'(b) - f'(a) \\ \implies \int_0^t f''(\tau) d\tau &= f'(t) - f'(0) \end{aligned}$$

Taking Laplace on both sides,

$$\begin{aligned} \mathcal{L} \left\{ \int_0^t f''(\tau) d\tau \right\} &= \mathcal{L} \{f'(t) - f'(0)\} \\ \implies \frac{1}{s} \mathcal{L} \{f''(t)\} (s) &= \mathcal{L} \{f'(t)\} - \mathcal{L} \{f'(0)\} \end{aligned}$$

Substituting our previous result for Laplace of a derivative here, we get...

$$\begin{aligned} \frac{1}{s} \mathcal{L} \{f''(t)\} (s) &= [s \cdot \mathcal{L} \{f(t)\} - f(0)] - \frac{f'(0)}{s} \\ \implies \mathcal{L} \{f''(t)\} (s) &= s^2 \cdot \mathcal{L} \{f(t)\} (s) - s \cdot f(0) - f'(0) \end{aligned}$$

2.7 Laplace Transform of Dirac-Delta Function

An impulse function is basically a function that takes a high value for a small interval such that the area under the curve remains constant = 1. It is denoted as

$$g(t)_{t_0}^\tau = \begin{cases} \frac{1}{2\tau}, & t \in (t_0 - \tau, t_0 + \tau) \\ 0, & t \notin (t_0 - \tau, t_0 + \tau) \end{cases}$$

The Dirac-Delta function is just a limiting case of the impulse function where this non-zero interval $2\tau \rightarrow 0$.

$$\delta(t - t_0) = \lim_{\tau \rightarrow 0} g(t)_{t_0}^\tau$$

And hence, would mathematically become,

$$\delta(t - t_0) = \begin{cases} \infty, & t = t_0 \\ 0, & t \neq t_0 \end{cases}$$

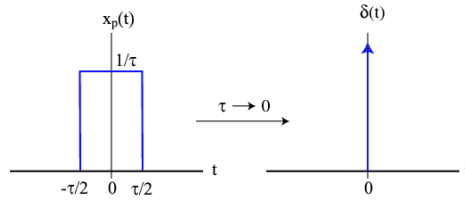


Figure 1: Dirac-Delta function as the limiting case of an Impulse function

Property of all Impulse functions (including Dirac-Delta Function)

Area under all impulse function = 1. Or in other words, the integral from $-\infty$ to $+\infty$ for all such impulse functions equals unity.

$$\int_{-\infty}^{\infty} g(t)_{t_0}^\tau dt = 1$$

$$\lim_{t \rightarrow t_0} \delta(t - t_0) = 0, \forall t \neq t_0$$

Laplace of any Impulse function is given by :

$$\begin{aligned}
\mathcal{L}\{g(t)_{t_0}^\tau\}(s) &= \int_0^\infty g(t)_{t_0}^\tau e^{-st} dt \\
&= \int_0^{t_0-\tau} 0 e^{-st} dt + \int_{t_0-\tau}^{t_0+\tau} \frac{1}{2\tau} e^{-st} dt + \int_{t_0+\tau}^\infty 0 e^{-st} dt \\
&= \frac{1}{2\tau} \int_{t_0-\tau}^{t_0+\tau} e^{-st} dt \\
&= \frac{1}{2\tau} \left[\frac{e^{-st}}{-s} \right]_{t_0-\tau}^{t_0+\tau} \\
&= \frac{e^{-st_0}}{2\tau s} (e^{\tau s} - e^{-\tau s})
\end{aligned}$$

Laplace of Dirac-Delta is obtained from this by :

We know that

$$\mathcal{L}\{\delta(t - t_0)\} = \mathcal{L}\{\lim_{\tau \rightarrow 0} g(t)_{t_0}^\tau\}$$

Since, Laplace Transform is continuous, I can swap the limit and Laplace transform.

$$\begin{aligned}
\mathcal{L}\{\delta(t - t_0)\} &= \lim_{\tau \rightarrow 0} (\mathcal{L}\{g(t)_{t_0}^\tau\}) \\
&= \lim_{\tau \rightarrow 0} \frac{e^{-st_0}}{2\tau s} (e^{\tau s} - e^{-\tau s}) \\
&= \frac{e^{-st_0}}{2s} \lim_{\tau \rightarrow 0} \left(\frac{e^{\tau s} - e^{-\tau s}}{\tau} \right)
\end{aligned}$$

Applying L'Hôpital's Rule,

$$\begin{aligned}
\mathcal{L}\{\delta(t - t_0)\} &= \frac{e^{-st_0}}{2s} \lim_{\tau \rightarrow 0} \left(\frac{se^{\tau s} + se^{-\tau s}}{1} \right) \\
&= \frac{e^{-st_0}}{2s} \cdot 2s = e^{-st_0}
\end{aligned}$$

Hence, Laplace Transform of the Dirac Delta function at t_0 is given by

$$\boxed{\mathcal{L}\{\delta(t - t_0)\} = e^{-st_0}}$$

This implies that $\mathcal{L}\{\delta(t)\}(s) = 1$. Hence obtaining the inverse Laplace Transform of 1 as $\mathcal{L}^{-1}\{1\}(t) = \delta(t)$

3 Solving ODE's using Laplace Transforms

Question: Solve the differential equation $y'' - 2y' + 5y = 8\sin(t) - 4\cos(t)$ if $y(0) = 1$ and $y'(0) = 3$.

Answer: Take Laplace Transform on both sides

$$\mathcal{L}\{y'' - 2y' + 5y\} = \mathcal{L}\{8\sin(t) - 4\cos(t)\}$$

$$\mathcal{L}\{y''\} - 2\mathcal{L}\{y'\} + 5\mathcal{L}\{y\} = 8\mathcal{L}\{\sin(t)\} - 4\mathcal{L}\{\cos(t)\}$$

Replacing $\mathcal{L}\{y''\}$ and $\mathcal{L}\{y'\}$ using properties, we get...

$$(s^2\mathcal{L}\{y\} - sf(0) - f'(0)) - 2(s\mathcal{L}\{y\} - f(0)) + 5\mathcal{L}\{y\} = 8\frac{1}{s^2 + 1} - 4\frac{s}{s^2 + 1}$$

$$(s^2 - 2s + 5)\mathcal{L}\{y\} - s - 1 = \frac{8 - 4s}{s^2 + 1}$$

Simplifying, we get

$$\begin{aligned}\mathcal{L}\{y\} &= \frac{1}{s^2 - 2s + 5} \left(\frac{8 - 4s}{s^2 + 1} + s + 1 \right) \\ \Rightarrow \mathcal{L}\{y\} &= \frac{s^3 + s^2 - 3s + 9}{(s^2 + 1)(s^2 - 2s + 5)} \\ \Rightarrow y &= \mathcal{L}^{-1} \left\{ \frac{s^3 + s^2 - 3s + 9}{(s^2 + 1)(s^2 - 2s + 5)} \right\}\end{aligned}$$

seperating using partial fractions, we get...

$$\begin{aligned}y &= \mathcal{L}^{-1} \left\{ \frac{2}{s^2 + 1} + \frac{s - 1}{s^2 - 2s + 5} \right\} \\ \Rightarrow y &= 2 \cdot \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 1} \right\} + \mathcal{L}^{-1} \left\{ \frac{(s - 1)}{(s - 1)^2 + 2^2} \right\} \\ \Rightarrow y &= 2\sin(t) + e^t \cos(2t)\end{aligned}$$

It is important to understand the last step. we know that $\mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 2^2} \right\} = \cos(2t)$. Shifting the Laplace Transformed function a unit to the right is caused by multiplying a function by e^{at} . Here, we have $\mathcal{L}^{-1} \left\{ \frac{(s-1)}{(s-1)^2 + 2^2} \right\}$ which means the Laplace Transformed function has been shifted 1 unit to the right.

Question: Solve the Differential Equation $2y'' + y' + 2y = \delta(t - 5)$ if $y(0) = 0$ and $y'(0) = 0$

Answer: Taking Laplace Transform of the given ODE on both sides,

$$\begin{aligned}\mathcal{L}\{2y'' + y' + 2y\} &= \mathcal{L}\{\delta(t - 5)\} \\ 2\mathcal{L}\{y''\} + \mathcal{L}\{y'\} + 2\mathcal{L}\{y\} &= \mathcal{L}\{\delta(t - 5)\}\end{aligned}$$

Substituting for $\mathcal{L}\{y''\}$, $\mathcal{L}\{y'\}$ and $\mathcal{L}\{\delta(t - 5)\}$ we get,

$$2s^2 \mathcal{L}\{y\} - 2s f(0) - 2f'(0) + s \mathcal{L}\{y\} - f(0) + 2 \mathcal{L}\{y\} = e^{-5s}$$

Substituting given values for $f(0)$ and $f'(0)$ and then simplifying, we get

$$\begin{aligned}(2s^2 + s + 2) \mathcal{L}\{y\} &= e^{-5s} \\ \Rightarrow \mathcal{L}\{y\} &= \frac{e^{-5s}}{2s^2 + s + 2} \\ \Rightarrow y &= \mathcal{L}^{-1} \left\{ \frac{e^{-5s}}{2s^2 + s + 2} \right\} \\ \Rightarrow y &= \frac{1}{2} \mathcal{L}^{-1} \left\{ \frac{e^{-5s}}{s^2 + \frac{s}{2} + 1} \right\} \\ \Rightarrow y &= \frac{1}{2} \mathcal{L}^{-1} \left\{ e^{-5s} \cdot \frac{1}{\left(s + \frac{1}{4}\right)^2 + \left(\frac{\sqrt{15}}{4}\right)^2} \right\} \\ \Rightarrow y &= \frac{2}{\sqrt{15}} \mathcal{L}^{-1} \left\{ e^{-5s} \cdot \frac{\frac{\sqrt{15}}{4}}{\left(s + \frac{1}{4}\right)^2 + \left(\frac{\sqrt{15}}{4}\right)^2} \right\}\end{aligned}$$

Now, since $e^{-5s} = \mathcal{L}\{\delta(t - 5)\}$ and the horrendous looking second term is the Laplace transform of $e^{-t/4} \cdot \sin\left(\frac{t\sqrt{15}}{4}\right)$, I can replace that inside the Inverse Laplace

$$y = \frac{2}{\sqrt{15}} \mathcal{L}^{-1} \left\{ \mathcal{L}\{\delta(t - 5)\} \cdot \mathcal{L} \left\{ e^{-t/4} \cdot \sin\left(\frac{t\sqrt{15}}{4}\right) \right\} \right\}$$

Since, $\mathcal{L}\{f\} \cdot \mathcal{L}\{g\} = \mathcal{L}\{f * g\}$

$$y = \frac{2}{\sqrt{15}} \mathcal{L}^{-1} \left\{ \mathcal{L} \left\{ \delta(t-5) * \left[e^{-t/4} \sin \left(\frac{t\sqrt{15}}{4} \right) \right] \right\} \right\}$$

$$\Rightarrow y = \frac{2}{\sqrt{15}} \left(\delta(t-5) * \left[e^{-t/4} \sin \left(\frac{t\sqrt{15}}{4} \right) \right] \right)$$

Technically this is the required function. But leaving it in this convoluted form feels incomplete. So I shall simplify it to the normal form. For this, I use the definition of convolution $\{f * g\}(t) = \int_0^t g(\tau) f(t-\tau) d\tau$. Let my Dirac-Delta function be the $g(t)$ and the other term be my $f(t)$

$$y = \frac{2}{\sqrt{15}} \int_0^t \delta(\tau-5) f(t-\tau) d\tau$$

Now let me show you something magical here, $\delta(t-5)$ is the Dirac-Delta function that takes a non-zero value in an infinitesimal interval around 5 and is zero everywhere else. Let this infinitesimal range be $(5-\epsilon, 5+\epsilon)$. I can split the integral (assuming $t > 5$) as follows

$$y = \frac{2}{\sqrt{15}} \left(\int_0^{5-\epsilon} \delta(\tau-5) f(t-\tau) d\tau + \int_{5-\epsilon}^{5+\epsilon} \delta(\tau-5) f(t-\tau) d\tau + \int_{5+\epsilon}^t \delta(\tau-5) f(t-\tau) d\tau \right)$$

Notice how the first and last integral becomes zero because of the Dirac-Delta, leaving us with just the middle integral.

$$y = \frac{2}{\sqrt{15}} \int_{5-\epsilon}^{5+\epsilon} \delta(\tau-5) f(t-\tau) d\tau$$

The real beauty lies in the realization that the range of τ in this integral is so small around 5 that the function $f(t-\tau)$ is a constant in this interval and has the value $f(t-5)$. Therefore it can be taken out of the integral.

$$y = \frac{2}{\sqrt{15}} f(t-5) \int_{5-\epsilon}^{5+\epsilon} \delta(\tau-5) d\tau$$

By definition, Since the area under Dirac-Delta function = 1, every $\delta(t-t_0)$ integral containing the active point t_0 becomes equal to unity. I also substitute back the function $f(t)$ here.

$$y = \frac{2}{\sqrt{15}} f(t-5)$$

$$y = \frac{2}{\sqrt{15}} e^{-(t-5)/4} \sin \left(\frac{(t-5)\sqrt{15}}{4} \right)$$

Remember that based on my earlier assumption, this is only true for $t > 5$. This is because, if $t < 5$, there wouldn't exist even an infinitesimal interval where the Dirac-Delta is active and the whole thing collapses to 0. I can Hence write the answer as

$$y = \begin{cases} \frac{2}{\sqrt{15}} e^{-(t-5)/4} \sin\left(\frac{(t-5)\sqrt{15}}{4}\right) , & t \geq 5 \\ 0 , & t < 5 \end{cases}$$

If I write this using Heaviside function instead of piecewise, then the final answer would be

$$y = \frac{2}{\sqrt{15}} u_5(t) \cdot e^{\frac{5-t}{4}} \cdot \sin\left(\frac{(t-5)\sqrt{15}}{4}\right)$$

You're Welcome, My fellow Comrades in Science...